

Section 0 – Title and contact Information

- Title of the proposed solution track: **Scaling Climate-Smart Mechanized Innovations to Optimize Rice–Wheat Systems for Higher Productivity, Profitability, and Resilience in Eastern India**
- Title of the AoW2 Flagship that the solution track aligns with: **Flagship 2.2: Catalyzing Mechanization and Postharvest Services through Agribusiness Growth**
- Lead Scientist(s) / Contact Person(s) / Center(s) :
 - Dr. Sudhanshu Singh, Agronomy, IRRI
 - Dr. Suryakanta Khandai, Mechanization and Postharvest, IRRI
 - Dr. Vikram Patil, Agricultural Economics (Value Chain and Entrepreneurship Development), IRRI
 - Dr. Virender Kumar, Agronomy, IRRI
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 - Dr. Panneerselvam Peramaiyan, Cropping System Agronomy, IRRI
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- Emails of the Lead Scientist(s) / Contact Person(s) / Center(s):
 - Technical: Sudhanshu Singh: s.singh@cgiar.org;
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- Key CGIAR Center Partners (in the case of multi-Center Eols): CIMMYT
- Technical link to other Science Programs – Sustainable Farming

Section 1 – Innovation Package, Value Proposition & Alignment

Purpose: To establish what the innovation is, who the customers and end-users are, why it matters, and how it connects to S4I's Flagships and Grand Challenges.

To catalyze a systemic shift toward climate-resilient intensification of the rice–wheat system through mechanization solutions in Indo Gangetic Plains (IGP) by deploying system approach of mechanized DSR followed by ZT early wheat, bundled with best-bet agronomy and suitable varieties/hybrids for system optimization. Smallholder men and women farmers and FPOs are the end-users, along with extension systems, implementing agencies and private sector. This is very much aligned with Flagship 2.2: Catalyzing Mechanization and Postharvest Services through Agribusiness Growth.

- **The problem to be solved:** Describe the challenge you are seeking to overcome through the proposed Scaling Solution track. What country / countries will the Solution Track operate in? Be sure to clarify who you are seeking to benefit with this work, with attention paid to gender and social inclusion. (~150 words)

The rice–wheat system in Eastern Uttar Pradesh (EUP) of eastern Indo-Gangetic Plains, spanning ~3 million ha, remains impoverished due to large yield gaps and frequent climate shocks. A key driver for low productivity is delayed planting—late rice harvests push wheat sowing beyond the optimal Nov 1–20 window, causing ~50 kg/ha/day yield loss from terminal heat-stress¹. Late-planted rice also from terminal-drought. More than 50% area is planted late, making timeliness the exception¹. Traditional puddled transplanted rice (PTR) exacerbates delays, while consuming excessive labor, water, and energy, and increasing GHG emissions^{2,3}. Mechanized

dry direct-seeded rice (DSR) and zero-till (ZT) wheat offer a transformative opportunity to optimize cropping calendars, enabling timely rice planting, advancing wheat sowing by 14-21 days, and enhancing productivity, resilience, and farmer income¹. However, adoption remains low due to limited awareness, weak service-provision network, and policy gaps. This track will scale these innovations through multi-stakeholder platforms to unlock system-level productivity, resilience, and rural livelihood gains.

- **Your scaling ambition:** Describe the ambition of your proposed Scaling Solution track. Which of the KPIs do you believe you will contribute most towards, and at what level of quantitative and qualitative impact? (~150 words)

We aim to catalyze a systemic shift toward climate-resilient intensification of the rice–wheat system through mechanization solutions in the Indo Gangetic Plains (IGP) by deploying a system approach of mechanized DSR followed by ZT early wheat, bundled with best-bet agronomy and suitable varieties/hybrids for system optimization. The scaling effort will target 21 districts, covering > 200,000 ha under DSR and ZT-wheat within three years, benefiting >1 million smallholder farmers. Expected system-level benefits include yield gains of >1.0 t/ha; income gains of INR 20000/ha (CSISA, unpublished); GHG emission reductions of 35–40%, and employment for rural youths. Through an established public–private innovation platform (Samriddih Dhan Network), we will strengthen the local agribusiness ecosystem by supporting 400+ Custom Hiring Centers (CHCs), Farmer Producer Organizations (FPOs), and >5,000 private rural entrepreneurs, ensuring demand creation and sustainable delivery of mechanized services and seeds of improved varieties. In addition, enabling policies and incentive mechanisms will be linked (e.g., carbon credits).

- **Innovation & Value Proposition** – Describe the innovation or bundle or package (maturity, novelty, suitability for scaling) and the benefits it delivers to institutional partners/customers and end-users (e.g., productivity, resilience, nutrition, risk reduction, institutional capacity). Can include linkages to other science programs that will collaborate and work on the innovation/bundle/package (~150 words)

The core innovation is mechanized solution (mechanized DSR followed by ZT wheat), climate-smart alternatives to PTR and conventional-till wheat for optimizing rice-wheat cropping calendar to boost productivity, profitability and climate resilience. Scaling will be delivered through a comprehensive innovation package addressing key bottlenecks. This includes strengthening mechanization and service delivery networks by engaging CHCs, private service providers (PSPs), and FPOs; optimizing cropping systems by integrating DSR with crop residue management and ZT wheat; and promoting DSR-fit, stress-tolerant rice varieties and hybrids supported by robust seed systems. Targeted awareness and training initiatives will drive behavioural change among machine operators, PSPs, and extension agents. Cluster-based demonstrations with incentive-linked adoption will further accelerate uptake, while carbon market linkages incentivize climate-smart practices. Strategic partnerships under the DSR network convened by UPCAR, World Bank-WRG, and IRRI provide strong institutional backing. Ultimately,

the approach delivers cost savings, higher yields, climate resilience, and market opportunities for farmers, governments, and agribusiness actors alike.

- **Customers & Alignment** – Identify the main institutional customers (ministries, donor programs, agribusinesses, lenders) and end-users (farmers, SMEs, cooperatives). Show how the track aligns to an AoW2 Flagship and Grand Challenge. (~100 words)

Institutional clients include the Department of Agriculture, Government of UP, State Agricultural Universities, KVKs, the State Seed Corporation, the World Bank - supported UP-AGREES Project Coordination Unit, and DSR network members. Partners will drive scaling of mechanized solutions combined with tailored best bet agronomy and varieties/hybrids as key authorizers and co-investors.

End users are smallholder men and women farmers and rural youth, supported by a scaling network of CHCs, FPOs, Women WSHGs, private service providers, and agri-entrepreneurs.

This aligns with Grand Challenge 2 and Flagship 2.2 of S4I Science Program advancing inclusive market-driven mechanization and post-harvest services through Agribusiness Growth.

- **Evidence of Impact** – Provide citations and evidence of both effectiveness (impact) and scalability in practice (uptake beyond pilots, adoption drivers). (~150 words)

Mechanized DSR delivers yields comparable to best-managed PTR or higher than late PTR, while reducing production costs by USD 150/ha, saving 80% labor, and increasing farmer income by USD 150/ha in eastern India (Singh et al. 2020; CSISA Report). DSR also conserves water, cuts GHG emissions, and raises net income across India^{4,5,6,7,8,9}. Importantly, DSR boosts subsequent wheat yields by ~9%, increasing system-level productivity⁵. ZT wheat adoption resulted in additional yield gain of 498 kg ha⁻¹ (19 %) and economic gain of USD 110/ha relative to conventional till wheat¹⁰. In EUP, UP-AGREES diagnostic reveal that >60% of farmers miss the optimal wheat sowing window due to late rice harvests. Integrating DSR with ZTW advances sowing by 14–21 days and raises system-level yields by ~1.0 t/ha (CSISA). Success hinges on socio-economic bundling with integrated crop management, timely services, suitable varieties, and strong awareness campaigns and enabling policies and incentive mechanisms.

Section 2 – Scaling Pathway, Sustainability, Partnerships & Risks

Purpose: To assess whether there is a credible pathway for the innovation to scale through public, private, or PPP delivery models — supported by sustainability plans, partner demand and commitments, financial leverage, and risk mitigation.

- **Scaling Pathway & Sustainability** – Describe how the innovation bundle and/or package will scale (public, private, PPP). Explain how the solution will remain viable at scale (revenues, public financing, PPP, IFI/donor support), with milestones or cost-recovery targets. (~250 words)

Project will adopt a multi-pathway strategy to deliver sustainable scaling in EUP:

Suitability Mapping and Precision Targeting: Remote sensing and GIS-based suitability mapping will identify late-planted areas and target them for mechanized DSR and ZT wheat. This evidence-based targeting ensures efficiency and maximizes impact.

Public-Platform Pathway: Interventions will be embedded in the ongoing UP-AGREES project and the state's extension systems, including the DoA, KVKs, and the State Seed Corporation. Large-scale DSR–ZT wheat clusters will be established in suitable late-planted areas, supported by state-backed incentives. Service providers will be linked to government mechanization schemes and seed subsidies, ensuring institutional coordination, mass outreach, and state-level validation.

Private-Service Pathway: A vibrant private-sector ecosystem will be fostered by strengthening entrepreneurship and service provision models through CHCs, private service providers, FPOs, and WSHGs. Partnerships with financial institutions will facilitate credit opportunities, while CSR co-investment will support scaling. This will increase access to machines and services on a rental basis to smallholders

Strategic Partnerships and Institutionalization: Guided by the IPSR framework, this initiative will be embedded in the MEL system of the World Bank-supported UP-AGREES loan project. Collaboration with Bayer, which is working with IRRI to scale DSR to 1 million ha in India, will provide strong leverage in EUP. Over time, public financing will taper, replaced by sustainable revenues from service fees, seed sales, and building a resilient market-driven ecosystem.

Communication and ME&L: Action-oriented communications will raise awareness and create demand through government and private advisory platforms. A robust ME&L system will track adoption constraints and support adaptive scaling.

- **Partnerships & Leverage** – Identify internal CGIAR champions and external partners that have a stake in championing the innovation bundle or package, provide evidence of demand (surveys, requests for collaboration, alignment with policies or strategies) and commitment (MOUs, LOIs, budget allocations from partners, or other evidence of demand), and specify major non-CGIAR programs or investments this track can leverage (budgets, partners, policies, programs, etc). (~300 words)

This initiative builds on the World Bank–funded UP-AGREES project, where IRRI serves as technical service provider and convenes multi-stakeholder partnerships. IRRI also leads the DSR Consortium with partners including Bayer, Corteva, BASF, Zydex, String Bio, Yara Fertilizers, ICAR, and several State Agricultural Universities. IRRI along with UP Government and World Bank-WRG also established Samridhi Dhan Network of public-private partners with focus to scale DSR in EUP. Bayer has committed to expand DSR to 1 million hectares in India, offering strong leverage for EUP.

- **Public Agencies:** The Department of Agriculture, Government of UP, the UP-AGREES PCU, ICAR-ATARI, and KVKs will drive policy alignment, extension integration, farmer mobilization, and co-delivery of training and demonstrations, ensuring institutional ownership and alignment with state priorities.
- **R4D Partners:** The World Bank provides the core investment platform through UP-AGREES. CGIAR programs such as Scaling for Impact & Sustainable farming and bilateral projects such as CSISA and IRRI-South Asia Regional Center funded by Govt of India, alongside SAUs, will contribute technical expertise, scaling science, and cross-regional learning.
- **Private Sector:** Bayer, Savannah Seeds, Corteva, IOCL and SEAL, machinery manufacturers and input dealers, local service providers, CHCs, are key partners for supporting sustainable market for seed, bio-residue and mechanizations
- **Farmer Collectives:** FPOs, WSHGs, and PSP networks will act as last-mile delivery agents offering mechanized solutions (DSR & ZT wheat), produce quality seeds, and manage residue aggregation, ensuring the innovation reaches smallholders effectively.

Leverage Strategy: Resources will be pooled across projects like UP-AGREES and CSISA, Direct-seeded Rice Consortium alongside state subsidies for mechanization, seed, and residue management. A joint MEL system will transparently attribute outcomes to co-investors, fostering accountability and long-term partnership commitments.

- **Risks & Bottlenecks** – Identify specific risks (financial, political, operational, social, or others) and bottlenecks (policy, institutions, finance, technology) in the specific context and place that could hinder scaling. Describe potential mitigation strategies to address these risks and bottlenecks. (~250 words)

A proactive risk management strategy, monitored through a quarterly updated risk log, will address potential bottlenecks to ensure responsible and effective scaling:

- **Technical risk and mitigation:** High weed pressure is a major constraint in DSR. A robust Integrated Weed Management (IWM) program will be implemented through service providers and farmer training, with demonstrations of best practices, including the safer and effective herbicides. Recently, herbicide-tolerant rice is commercialized which will overcome weed control challenge in DSR and facilitates its wide-scale adoption
- **Institutional risk and mitigation:** Weak governance and business capacity in emerging CHCs and FPOs can hinder service quality and financial sustainability. To mitigate this, a business incubation component, offering mentorship in financial planning, operational management, and transparent governance frameworks.
- **Market risk and mitigation:** A mismatch between machinery supply and rental demand could lead to underutilization of assets and reduce profitability. A cluster-based aggregation model will be used to forecast demand accurately before deploying machinery supported by Cluster demonstrations for creating demands for these technologies. Service offerings will be diversified to ensure year-round revenue. As a part of this, hands-on training programs to create local repair and service facilities.

- **Social risk and mitigation:** Mechanization benefits could be captured by larger farmers, excluding women, youth, and marginalized groups. We will implement equity-focused targeting, with explicit quotas and incentives for establishing women- and youth-led enterprises. Participatory MEL will track benefit distribution to ensure
- **Climate risk and mitigation:** Drought or floods may affect DSR crop establishment and yields. Stress-Tolerant Rice Varieties and insurance linkages will buffer farmers against climate shocks.
- **Policy risk and mitigation:** Delays in aligning government schemes could stall incentives. Formal MoUs and a high-level steering committee with key departments will ensure streamlined coordination and timely delivery.

Section 3 – Responsible Scaling Trajectory & Achievable Scaling Target

Purpose: To define what can realistically be achieved with S4I support in the next 2–3 years — including achievable reach, milestones, resources, activities, and safeguards for responsible scaling.

- **Readiness & Responsible Scaling** – Provide evidence of scaling readiness (IPSR or similar evidence). Describe actions for ensuring that scaling will be inclusive (extending to women, youth, marginalized groups) and environmentally sustainable. (*~150 words*)

The DSR innovation package demonstrates high scaling readiness, backed by validated technical efficacy, committed partners, and proven farmer interest. The innovations have been successfully piloted in EUP and scaled in similar agro-ecologies like Odisha, Bihar, Punjab, providing a robust evidence base. Applying IPSR framework, we find that readiness gaps lie not in the core technology but in the enabling environment—particularly service delivery networks and policy incentives. This proposal directly addresses those gaps to accelerate adoption.

Responsible scaling is central to the initiative and will be operationalized through a multi-pronged approach:

- **Equity-Focused Targeting:** Prioritizing WSHGs, youth, and marginalized communities for enterprise development and training to ensure inclusivity
- **Participatory MEL:** Continuously monitoring for unintended consequences (e.g., labor displacement, herbicide overuse) through feedback loops with communities.
- **Environmental and Social Safeguards:** Embedding GESI principles and environmental best practices into all training modules and partnership agreements.
- **Adaptive Management:** Utilizing annual stage-gate reviews to assess progress and socio-environmental impacts, with clear triggers to pivot or halt activities if negative spillovers arise.
- **Achievable target, Resources & Activities** – Define what can realistically be achieved in 2–3 years starting 2026 forward with S4I support and current/planned partnerships.

Outline the key resources (staff, institutional, financial, technical) and activities (e.g., training, policy engagement, distribution) required. (~250 words)

Over a 4-year timeframe, this initiative will achieve the following tangible targets:

Targets:

- **~200,000 ha** under DSR fb ZT wheat and DSR-fit varieties benefiting **> 1 million** smallholders' farmers.
- **400+** functional CHCs/PSPs established or strengthened
- **10,000** stakeholders trained (farmers, operators, extensionists)
- **4,000-5,000** new rural enterprises created (seed, nursery, services)

Resources: The initiative will be primarily resourced by leveraging UP-AGREES project funds (~55.35 M USD USD), and IRRI-South Asia Regional centre in UP (15 million USD) with state government subsidies for mechanization and seeds, and co-financing from private sector (CSR) and donor partners. Human resources will include dedicated extension staff from the DoA, scientists from IRRI & CIMMYT, trainers from KVKs, and business mentors from the private sector.

Activities:

- **Suitability mapping, diagnostics and precision targeting:** Conduct suitability mapping for DSR and late planted wheat areas for precision targeting of activities and cluster demonstrations
- **Large scale cluster Demonstrations of DSR and ZT wheat** with tailored best agronomy and varieties/hybrids through public-private partnership networks established under UPAGREES to catalyse scaling.
- **Strengthening service provision business model** to enhance access of capital-intensive mechanized technologies on rental basis to smallholder men and women farmers
- **Strengthening DSR-Fit Varieties and Seed Systems through** decentralized seed production networks through FPOs and WSHGs, private seed production entrepreneurship to ensure accessibility and availability of good quality certified seeds.
- **Capacity Building of value chain actors**
- **Agribusiness Incubation:** Provide business development support, financial linkages, and mentorship to CHCs and FPOs, and potential service providers.
- **Awareness and Outreach:** using mass media, digital platforms, and field days.
- **Policy and Partnerships:** Advocate for DSR cluster incentives and formalize PPPs for residue-based bioenergy value chains.
- **Carbon credit:** Linking DSR and zero-till wheat clusters with carbon credit programs or government schemes to incentivize adoption.

Timeline, Milestones & Monitoring – Provide an estimate of annual milestones for 2026–2028 (users reached, partnerships formed, policies adopted). Explain how progress will be monitored, lessons captured, and adaptations made. (~200 words)

Year	Milestones	Monitoring
2026 – Foundation & Piloting	Complete baseline diagnostics and suitability mapping for DSR and ZT wheat for 21 districts; sign MoUs with partners; 10,000 ha under DSR fb ZT wheat and DSR-fit varieties; introduce crop residue management technologies through CHCs/PSPs; launch community seed pilots via FPOs/WSHGs; initiate training for seed entrepreneurs, operators, and extension agents.	Track partner onboarding, clusters formed, trainings delivered, and early farmer feedback.
2027 – Acceleration & Expansion	30,000 ha under DSR fb ZT wheat and DSR-fit varieties; operationalize 200 CHCs/PSPs with business plans and financing; scale decentralized STRV seed networks and community seed banks; launch awareness campaigns; establish PPPs for bio-residue value chains with IOCL/SEAL; conduct first IPSR scaling readiness review.	Assess seed availability, adoption rates, CHC uptake, revenue, and farmer satisfaction.
2028 – Consolidation & Institutionalization	70,000 ha under DSR fb ZT wheat and DSR-fit varieties; expand CHCs/PSPs to 400; institutionalize STRV seed chains within UP-AGREES/KVK plans; mainstream DSR into state programs; conduct mid-term impact assessment.	Track institutionalization of seed systems, CHC/service network expansion, and CGIAR scaling indicators.
2029 – Sustainability & Handover	90,000 ha under DSR fb ZT wheat and DSR-fit varieties; transition CHC/FPO/seed networks to federations; publish a DSR Scaling Playbook; conduct final sustainability and impact evaluation.	Evaluate financial sustainability of CHCs, FPOs, and seed networks; assess institutional capacity for independent scaling.

Monitoring Framework: The MEL system will be embedded within the UP-AGREES framework, customized to include CGIAR's key scaling indicators. Annual participatory scaling readiness assessments, based on the IPSR methodology, will be a cornerstone of the M&E plan, enabling the project team to identify and address emerging bottlenecks and adaptively manage the scaling strategy for maximum impact.

Annexes (see attached PDF files)

- IPSR Innovation Profile (if available): Attached as an enclosure
- Scaling Readiness assessment results

Readiness score is **9**. The innovation is validated for its ability to achieve a specific impact under uncontrolled conditions. (The scaling readiness assessment was done using: [Calculator - Readiness | Scaling readiness](#))

- Scaling Scan results (if available)

Scaling Scan results is 3.5. The innovation is validated for its ability to achieve a specific impact under uncontrolled conditions. (The scaling scan results was generated using <https://www.scalingscan.org/>)

- Prior project reports
- Media or other evidence of prior scaling work that can be accelerated

<https://www.thehindu.com/news/national/uttar-pradesh/up-cm-yogi-adityanath-unveils-two-initiatives-with-support-of-world-bank/article69558436.ece>

- MELIA studies, impact assessments, reports, publications
- Partner commitments (MOUs, LOIs, letters of support)
 - <https://csisa.org/wp-content/uploads/sites/2/2014/06/CSISA-Phase-III-Annual-Report-2016.pdf>
 - Government of Uttar Pradesh approved the 6-year project (in the process of signing the agreement) to scale the DSR-ZT wheat systems through above mentioned interventions and technologies.
 - IRRI South Asia Regional Center (ISARC) has made MoUs/LoIs with SAUs, ICAR institutions, and private sector
 - MoU: ANDUAT, Kumarganj, BUAT Banda, CSAUAT Kanpur, SVPUAT Modipuram [<https://www.irri.org/news-and-events/news/irri-inks-mou-uttar-pradesh-agricultural-universities>] SHUATS Prayagraj, MGUG Gorakhpur [].
 - LOI: ICAR-NBAIM [<https://www.irri.org/news-and-events/news/irri-nbaim-advance-microbial-innovations-improved-rice-systems-india>], ICAR-NBPGR [<https://www.irri.org/news-and-events/news/irri-and-icar-further-promote-research-and-development-rice-genetic-resources>]
 - Deed of Donation: ALCOM for Pyrolysis and production of Biochar